



Provider Network Adequacy Analysis

Evaluating whether a payer's provider network meets access standards across specialty, distance, and wait-time thresholds | Sohail — Strategic HealthCare BI Analyst

672

Network Providers
Across 10 Specialties

48,000

Synthetic Members
Analyzed

78.3%

Fully Compliant
(All 3 Standards)

99.7%

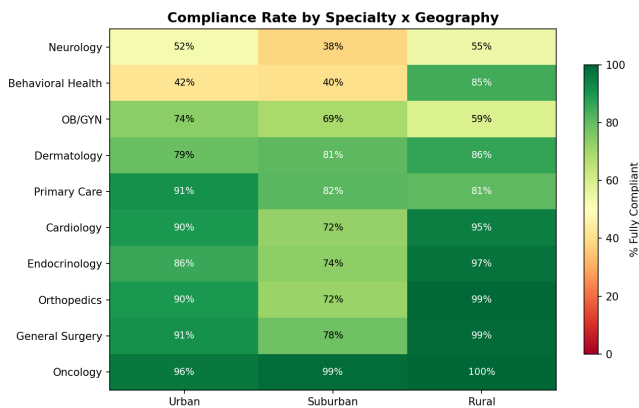
Distance
Compliance

78.6%

Wait-Time
Compliance

Headline finding: The network looks adequate on paper but isn't adequate in practice. It meets provider-to-member ratio standards almost universally (100%) and distance standards nearly everywhere (99.7%) — yet appointment wait-time compliance falls to 78.6%, driven by Neurology and Behavioral Health backlogs of 22–46 days. Headcount on the roster isn't the same as capacity a member can actually access.

Compliance by Specialty x Geography



Where the Network Breaks Down

- **Neurology & Behavioral Health** fail across nearly every county (19–43% compliant), driven by wait times of 22–46 days — not distance.
- **Suburban compliance (70.5%) is worse than rural (85.7%)** — a counterintuitive result. Dense suburban demand overwhelms a thin specialist panel more than sparse rural demand does.
- **OB/GYN access lags in rural counties** (Millbrook, Cedar Ridge) — a concern given the acuity of delayed prenatal care.
- Every specialty passes the **provider-to-member ratio** standard, confirming the gap is an appointment-availability problem, not a network-size problem.

Methodology

Synthetic network of 672 providers and 48,000 members across 8 counties (urban/suburban/rural). Haversine distance to nearest accepting in-network provider, simulated next-available-appointment wait time, and provider-to-member ratio were each compared to a benchmark table modeled on CMS Medicare Advantage Time & Distance criteria and state Medicaid MCO wait-time standards.

Recommended Actions

- Prioritize Behavioral Health / Neurology telehealth expansion in the worst-performing counties — closes the wait-time gap without new physical sites.
- Pair ratio compliance with an appointment-availability audit before every regulatory filing.
- Model specialist demand at the county level; the suburban/rural pattern shows urban-suburban-rural buckets can mask real gaps.

Overall Compliance by Geography

